

Package ‘dsHarmonization’

February 17, 2026

Title What the Package Does (One Line, Title Case)

Version 0.0.0.9000

Description What the package does (one paragraph).

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cast_to_factorDS	<i>Cast specified columns to factor</i>
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Description

Convert one or more columns of a data frame to factor type.

Usage

```
cast_to_factorDS(df, columns)
```

Arguments

df	A data frame containing the columns to be converted.
columns	A single character string specifying the column names separated by "\$".

Value

The input data frame with the specified columns converted to factors.

check_categoricalDS	<i>Check validity of categorical variables</i>
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Description

Verify that selected categorical variables in a data frame contain only predefined valid values.

Usage

```
check_categoricalDS(df)
```

Arguments

df	A data frame containing the categorical variables to be checked.
----	--

Value

A character vector with the names of columns containing invalid values. Returns an empty character vector if all checked variables are valid.

check_missing_dataDS	<i>Compute percentage of missing values per column</i>
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Description

Computes the proportion of missing values (NA) for each column of a data frame. The function returns a named numeric vector containing the percentage of missing values for every variable.

Usage

```
check_missing_dataDS(df)
```

Arguments

df	A data frame for which the percentage of missing values per column is to be calculated.
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Value

A named numeric vector where each element represents the percentage of missing values in the corresponding column of df.

check_numericDS	<i>Check validity of numeric variables</i>
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Description

Verify that selected numeric variables in a data frame fall within predefined ranges and satisfy integer constraints where required. Also checks the format of the patient identifier (pat_ID), if present.

Usage

```
check_numericDS(df)
```

Arguments

df	A data frame containing the numeric variables to be checked.
----	--

Value

A character vector with the names of columns containing invalid values. Returns an empty character vector if all checked variables are valid.

check_patient_visitDS	<i>Count patient-visit combinations</i>
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Description

Create and count all patient-visit combinations in a data frame, grouping by patient identifier and visit time from diagnosis.

Usage

```
check_patient_visitDS(df)
```

Arguments

df	A data frame containing at least the columns pat_ID and Visit_months_from_diagnosis.
----	--

Value

A data frame with one row per patient-visit combination and a column n indicating the number of occurrences.

check_variablesDS	<i>Check that required variables are present in a data frame</i>
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Description

Verify that a data frame contains the expected variables.

Usage

```
check_variablesDS(df, variables_string)
```

Arguments

df	A data frame to be checked.
variables_string	A single character string specifying the expected variable names separated by "\$".

Value

A named list with two elements:

- missing: variables specified but not found in the data frame.
- extra: variables found in the data frame but not specified.

fill_missing_dataDS	<i>Fill missing values by patient and visit order</i>
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Description

For each patient, sort visits by time from diagnosis and propagate non-missing values of a specified column. Missing values are replaced with the most recent non-missing value observed in chronological order within the same patient.

Usage

```
fill_missing_dataDS(df, pat_id_col, visit_col, value_col)
```

Arguments

df	A data frame containing at least the patient identifier column, the visit time column, and the variable to be filled.
pat_id_col	A character string specifying the name of the patient identifier column.
visit_col	A character string specifying the name of the visit time column.
value_col	A character string specifying the name of the column whose missing values should be filled.

Value

A data frame where missing values in value_col are filled forward within each patient group according to increasing visit time.

global_imputationDS	<i>Global imputation of missing data using MICE</i>
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Description

Perform a global imputation of missing values in a data frame using the Multiple Imputation by Chained Equations (MICE) algorithm. The function automatically identifies continuous and categorical variables, assigns appropriate imputation methods (pmm, logreg, polyreg), and returns a single imputed data frame.

Usage

```
global_imputationDS(df, id_col, m, maxit)
```

Arguments

df	A data frame containing the variables to be imputed. Both numeric and categorical variables are supported.
id_col	A string specifying the name of the identifier column. This column is excluded from the imputation process and used to aggregate the results across multiple imputations.
m	Number of imputations to generate (passed to mice).
maxit	Maximum number of MICE iterations per imputation chain.

Value

A data frame where missing values have been imputed using the MICE

remove_columnsDS	<i>Remove specified columns from a data frame</i>
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Description

Remove one or more columns from a data frame based on their names.

Usage

```
remove_columnsDS(df, col_names)
```

Arguments

df	A data frame from which columns will be removed.
col_names	A character vector specifying the names of the columns to be removed.

Value

The input data frame with the specified columns removed.

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